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SYSTEMS AND METHODS FOR IMPROVED SPATIAL COMPUTING SYSTEMS

Abstract

A method for improved heat discharge may include sensing a temperature value of a battery circuit; activating a heat dissipation element within the battery circuit when the temperature value reaches a threshold; discharging heat from the battery circuit via the activated heat dissipation element; and deactivating the heat dissipation element when the temperature value falls below the threshold.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority to U.S. Application 63/560,955, filed 4 Mar. 2024, U.S. Application 63/567,132, filed 19 Mar. 2024, U.S. Application 63/566,823, filed 18 Mar. 2024, U.S. Application 63/671,294, filed 15 Jul. 2024, and U.S. Application 63/687,888, filed 28 Aug. 2024, the disclosures of each of which are incorporated, in their entirety, by this reference.

BRIEF DESCRIPTION OF DRAWINGS
[0002] The accompanying drawings illustrate a number of exemplary embodiments and are a part of the specification. Together with the following description, these drawings demonstrate and explain various principles of the present disclosure.
[0003] FIG. **1** is an illustration of an example battery protection circuit according to some embodiments of this disclosure.
[0004] FIG. **2** is an illustration of an example battery protection device according to some embodiments of this disclosure.
[0005] FIG. **3** is an illustration of an exemplary head-mounted display that is configured to provide both a real-world view and a virtual world view to a user, according to some embodiments of this disclosure.
[0006] FIG. **4** is an illustration of an example differentially private retrieval-augmented diffusion model images, according to some embodiments of this disclosure.
[0007] FIG. **5** is an illustration of an example differentially private retrieval-augmented diffusion model pipeline, according to some embodiments of this disclosure.
[0008] FIG. **6** is an illustration of an example differentially private retrieval-augmented diffusion model retrieval pipeline, according to some embodiments of this disclosure.
[0009] FIG. **7** is an illustration of an example differentially private retrieval-augmented diffusion model Shutterstock samples, according to some embodiments of this disclosure.
[0010] FIG. **8** is an illustration of an example differentially private retrieval-augmented diffusion model samples, according to some embodiments of this disclosure.
[0011] FIG. **9** is an illustration of example adaptive tessellation quads, according to some embodiments of this disclosure.
[0012] FIG. **10** is an illustration of example 5-bit value quads, according to some embodiments of this disclosure.
[0013] FIG. **11** is an illustration of an example adaptive tessellation quads representing a z-surface, according to some embodiments of this disclosure.
[0014] FIG. **12** is an illustration of an example modified adaptive tessellation quads representing a

z-surface, according to some embodiments of this disclosure.
[0015] FIG. **13** is an illustration of an example artificial-reality system according to some embodiments of this disclosure.
[0016] FIG. **14** is an illustration of an example artificial-reality system with a handheld device according to some embodiments of this disclosure.
[0017] FIG. **15**A is an illustration of example user interactions within an artificial-reality system according to some embodiments of this disclosure.
[0018] FIG. **15**B is an illustration of example user interactions within an artificial-reality system according to some embodiments of this disclosure.
[0019] FIG. **16**A is an illustration of example user interactions within an artificial-reality system according to some embodiments of this disclosure.
[0020] FIG. **16**B is an illustration of example user interactions within an artificial-reality system according to some embodiments of this disclosure.
[0021] FIG. **17** is an illustration of an example wrist-wearable device of an artificial-reality system according to some embodiments of this disclosure.
[0022] FIG. **18** is an illustration of an example wearable artificial-reality system according to some embodiments of this disclosure.
[0023] FIG. **19** is an illustration of an example augmented-reality system according to some embodiments of this disclosure.
[0024] FIG. **20**A is an illustration of an example virtual-reality system according to some embodiments of this disclosure.
[0025] FIG. **20**B is an illustration of another perspective of the virtual-reality systems shown in FIG. **20**A.
[0026] FIG. **21** is a block diagram showing system components of example artificial- and virtual-reality systems.

Description

[0027] Throughout the appendices, identical reference characters and descriptions indicate similar, but not necessarily identical, elements. While the exemplary embodiments described herein are susceptible to various modifications and alternative forms, specific embodiments have been shown by way of example in the appendices and will be described in detail herein. However, the exemplary embodiments described herein are not intended to be limited to the particular forms disclosed. Rather, the present disclosure covers all modifications, equivalents, and alternatives falling within this disclosure.
DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENTS
Systems and Methods for Improved Heat Discharge
[0028] The present disclosure is generally directed to systems and methods for actively discharging a smart glasses case when certain temperature thresholds are met to avoid thermal damage. The idea uses a combination of hardware and firmware to detect when the case is at or approaching high temperatures using a battery temperature sensor and then algorithmically determine when it should start to actively discharge. An active discharge circuit can be implemented using several metal-oxide-semiconductor field-effect transistors (MOSFETs) with general-purpose input/output (GPIO) control, each connected to a high-power resistor. These MOSFETs can be turned on independently to control how much current is being discharged. The total current can be up to hundreds of mA. Splitting up the circuit into multiple parallel resistors also spreads out the heat dissipation in the case to avoid a specific hotspot that could damage components or the user.
[0029] The disclosed systems and methods for improved heat discharge may include a microcontroller that may sense a temperature value of a battery circuit using a thermal sensor (e.g.,

negative temperature coefficient thermistor (NTC)). For examples, FIG. **1** shows an example battery protection circuit according to some embodiments of this disclosure. The thermal sensor may be built into a battery pack within the battery circuit. In some methods, the microcontroller may activate a heat dissipation element within the battery circuit when the temperature value reaches a predetermined threshold. The microcontroller may connect to an active discharge circuit including heat dissipation elements, which may include but are not limited to high-power resistors, GPIO control units, and MOSFETS. The microcontroller may have independent control over each high-power resistor via a GPIO control over a MOSFET. The MOSFETS may be turned on independently to control the amount of current that may be discharged from the battery. The active discharge circuit may be split up into multiple parallel resistors that may spread out the heat dissipation and avoid hot spots that may damage surrounding components within the battery circuit. In other methods, the microcontroller may discharge heat from the battery circuit via the activated heat dissipation elements. The microcontroller may predict via firmware (e.g., algorithm, prediction model, etc.) when it is optimal to discharge heat from the battery depending on a variety of conditions, including when the temperature value reaches the predetermined threshold and when the battery reaches a safe level of voltage. In further embodiments, the microcontroller may deactivate the heat dissipation element when the temperature value falls below the predetermined threshold.

[0030] In some embodiments, the battery circuit may have various configurations and/or components. For example, FIG. **2** shows an example battery circuit according to some embodiments of this disclosure, including an active discharge circuit. In some examples, the active discharge circuit and additional components (including, e.g., an integrated circuit) may be spatially positioned away from the battery to minimize risk of thermal damage to key components and the battery.

[0031] The process parameters and sequence of the steps described and/or illustrated herein are given by way of example only and can be varied as desired. For example, while the steps illustrated and/or described herein may be shown or discussed in a particular order, these steps do not necessarily need to be performed in the order illustrated or discussed. The various exemplary methods described and/or illustrated herein may also omit one or more of the steps described or illustrated herein or include additional steps in addition to those disclosed.

[0032] The preceding description has been provided to enable others skilled in the art to best utilize various aspects of the exemplary embodiments disclosed herein. This exemplary description is not intended to be exhaustive or to be limited to any precise form disclosed. Many modifications and variations are possible without departing from the spirit and scope of the present disclosure. The embodiments disclosed herein should be considered in all respects illustrative and not restrictive. Reference should be made to any claims appended hereto and their equivalents in determining the scope of the present disclosure.

[0033] Unless otherwise noted, the terms “connected to” and “coupled to” (and their derivatives), as used in the specification and/or claims, are to be construed as permitting both direct and indirect (i.e., via other elements or components) connection. In addition, the terms “a” or “an,” as used in the specification and/or claims, are to be construed as meaning “at least one of.” Finally, for ease of use, the terms “including” and “having” (and their derivatives), as used in the specification and/or claims, are interchangeable with and have the same meaning as the word “comprising.”

Example Embodiments

[0034] Example 1: A method including (1) sensing a temperature value of a battery circuit; (2) activating a heat dissipation element within the battery circuit when the temperature value reaches a threshold; (3) discharging heat from the battery circuit via the activated heat dissipation element; and (4) deactivating the heat dissipation element when the temperature value falls below the threshold.

Bifocal AR/VR Headset for Business Use

[0035] Virtual-reality (VR) devices, augmented reality devices, and other artificial reality devices (collectively referred to as artificial-reality devices) can provide a rich, immersive, experience that enables users to interact with virtual objects and/or real objects that have been virtually augmented in some fashion. While artificial-reality devices are often utilized for gaming and other entertainment purposes, they are also commonly employed for purposes outside of recreation. For example, governments may use them for military training simulation, doctors may use them to practice surgery, and engineers may use them as visualization aids.
[0036] One example of an artificial-reality device is a head-mounted display (HMD) that fully immerses a user in a VR or other alternate reality experience. Conventional HMDs like this typically include a display housing that, when worn, prevents light from the user's external environment from entering the display housing, and thus, the user's field of view. While such a configuration may enhance the user's VR experience, this housing also prevents the user from viewing the real-world environment, which may make it difficult for the user to interact with real-world objects (including objects that are displayed and/or augmented in some fashion in VR). For example, a user sitting at a desk and wearing an HMD may find it difficult to operate a computer keyboard, a mouse, a stylus, or the like since the HMD (and, in particular, the HMD's display housing) blocks the user's view of such objects.
[0037] The present disclosure is generally directed to an HMD device configured to provide a user with a substantially unobstructed view, through a bottom portion of the device's housing, of the user's real-world environment. As will be explained in greater detail below, an HMD device may include (1) a display unit configured to display computer-generated imagery to a user and (2) a housing that retains the display unit. The HMD device may be mounted on the user's head and the display unit may be positioned in a top portion of a field of view of the user. The display unit may be dimensioned to provide the user with a substantially unobstructed bottom view of the field of view of a real-world environment to the user.
[0038] Features from any of the embodiments described herein may be used in combination with one another in accordance with the general principles described herein. These and other embodiments, features, and advantages will be more fully understood upon reading the following detailed description in conjunction with the accompanying drawings and claims. The following will provide, with reference to FIG. **3**, detailed descriptions of an HMD device worn by a user configured to provide a real-world view and a virtual world view.
[0039] Turning to FIG. **3**, exemplary HMD device **300** may both present a real-world view **106** and a virtual world view **304** to a user **302**. As used herein, an HMD device may generally refer to a device that is used to operate in a VR environment, an augmented reality environment, a mixed reality environment, or some combination thereof. HMD device **300** may present a variety of content to a user, including virtual views of an artificially rendered virtual-world view **304** and/or augmented view of real-world view **306**. HMD device **300** may include a display unit configured to display computer-generated imagery to user **302** and a housing that retains the display unit.
[0040] As illustrated in FIG. **3**, HMD device **300** may be adapted to combine real-world view **306** and virtual world view **304** for user **302**. For example, HMD device **300** may be mounted on a head of user **302** such that the display unit, which may be configured to display a virtual world view **304**, is positioned in a top portion of the field of view of the user **302**. While the display unit may obstruct the top portion of the field of view of the user **302**, the housing of HMD device **300** may be dimensioned to provide the user with a substantially unobstructed bottom portion of the field of view of the real-world view **306** to the user **302**. In some embodiments, HMD device **300** may completely obstruct the user's view of the real world **306**. In other embodiments, HMD device **300** may only partially obstruct the user's view of the real world **306** and/or may obstruct the field of view of user **302** depending on content being displayed on the display unit of HMD device **300**.
[0041] In some embodiments, an alternative face cushion (not shown in drawings) that defines an opening around a nose area of user **302** may be used to provide an unobstructed view of the real-

world view **306** to the user **302**. In some embodiments, lenses within the HMD device **300** may be shifted (relative to traditional orientations) since the eyes of user **302** will be located lower relative to the display unit. In addition, software may be modified to center virtual images generated by the display unit at a new, raised height since the display housing will not cover the entire field of view of user **302**.

[0042] Because HMD device **300** may be adapted to provide a user with both real-world view **306** and virtual world view **304**, HMD device **300** may obviate the need for virtual keyboards and/or other artifacts that may be unsuitable for practical use. For example, the exemplary HMD configuration described and illustrated herein may enable a user to quickly and seamlessly interact with both real-world objects (e.g., physical keyboards) and virtual objects (displayed, e.g., on a display unit of an HMD) without having to don or doff the HMD to do so.

Example Embodiments

[0043] Example 2: A head-mounted display including (i) a display unit configured to display computer-generated imagery to a user and (ii) a housing that retains the display unit, where, when the head-mounted display is mounted on a head of the user and the display unit is positioned in a top portion of a field of view of the user, the display unit obstructs the top portion of the field of view of the user and the housing is dimensioned to provide the user with a substantially unobstructed bottom portion of the field of view of a real-world environment to the user.

Systems and Methods for Improved Diffusion Models

[0044] The present disclosure is generally directed to systems and methods for improved diffusion models that employ a novel technique to train a high-quality text-to-image generative model by using two sets of training data. This training data includes a “public” dataset of images for which no guarantee of privacy is provided, such as data that has been purchased or licensed with rights to reproduce, and a “private” dataset of images that the model should not memorize or be capable of reproducing exactly. At a high level, the present disclosure employs a novel way of combining differential privacy guarantees with retrieval-augmentation for image generation. This disclosure may also inherit all the benefits of retrieval-augmented generative models. Additionally, it may be possible to easily remove or add data from/to the retrieval database after the model has been trained, without needing to do any retraining. Further, this disclosure may negate the computational limitations of traditional differential privacy methods and may enable the use of large-scale, high-fidelity image generative models with differential privacy guarantee.

[0045] In one example, the disclosed systems and methods for improved diffusion models may include an architecture for a differentially private retrieval-augmented latent diffusion model (DP-RDM) for image generation. For example, FIG. **4** shows an example differentially private retrieval-augmented diffusion generated images. The diffusion model may include a process for retrieval augmentation that incorporates differential privacy. In other embodiments, differential privacy guarantees may be obtained for data used in the retrieval dataset at inference time, where the use of differentially private stochastic gradient descent (DP-SGD) for training may be avoided. In some methods, the retrieval augmentation mechanism may include retrieval dataset subsampling, where the model is presented with a new prompt to generate images. For example, FIG. **5** shows an example differentially private retrieval-augmented diffusion model pipeline for generating the images. The retrieval data set may include a subset that is randomly selected and considered for responses to the prompt.

[0046] In some embodiments, the retrieval augmentation mechanism may include an aggregation method where the nearest neighbors (e.g., retrieval neighbors “k”) are selected among data sets (e.g., D and D′), where distance may be measured in an embedding space used by the model. For example, FIG. **6** shows an example differentially private retrieval-augmented diffusion model retrieval pipeline to apply the aggregation method. In further embodiments, the model may include a text encoder and an image encoder where the embedding spaces may be aligned. The prompt may be embedded using the text encoder, and the images in the retrieval dataset may be embedded via the image encoder. In some methods, after embedding a text prompt (e.g., user input), the nearest neighbors in the data sets may be retrieved and averaged.

[0047] In some methods, Isotropic Gaussian noise may be added to the aggregated embedding. In other embodiments, the level of privacy guaranteed provided by adding Isotropic Gaussian noise may be adjusted by changing the subsampling ratio used (e.g., smaller ratio=more private), the number of nearest neighbors aggregated (e.g., more neighbors=more private), and the variance of the noise added (e.g., higher variance=more private). In other methods, the precise level of privacy protection may be quantified using privacy accounting frameworks (e.g., Renyi Differential Privacy and Gaussian Differential Privacy). In further methods, the model may be trained using a public dataset, both for updating the weights, and as the retrieval dataset. After the training is completed, a private dataset may be added to the retrieval dataset. The trained data sets may be instantiated and demonstrated for the retrieval-augmented diffusion model for generating images (e.g., generated images shown in FIG. **7** and FIG. **8**). Additionally, the method may extend to models for generating text, further including multi-modal models (e.g., text+images, speech, music, audio, video, 3d meshes).

Example Embodiments

[0048] Example 3: A computer-implemented method of augmenting image data, the method including: (1) retrieving a subset of image data from a private database in response to a user input; (2) privatizing the subset of image data; and (3) generating at least one image based on the privatized subset of image data and the user input.

Systems and Methods for Adaptive Tessellation

[0049] The present disclosure is generally directed to a novel technique for adaptive tessellation that may convert a depth buffer into triangles that represent a z-surface. An algorithm may be used for converting a depth buffer into triangles. The algorithm may operate with a hardware implementation, where the algorithm operates inside a hardware unit (e.g., tile-based renderer) that may operate as a scene is being rendered. The algorithm may allow an error term to be specified which indirectly controls the number of triangles that are output. The algorithm may include two parts such as (1) deciding which depth values to ignore, and (2) computing the triangle topology for the depth values that are left. The hardware implementation inside the hardware unit may write out a data structure with each tile (e.g., 32×32 pixels), which may describe the triangles used to represent the z-surface. This data structure may be processed by software into a triangle list or may be processed directly by the hardware in some implementations. This may be accomplished by post processing the depth buffer after each tile is rendered.

[0050] In one example, the disclosed systems and methods for improved adaptive tessellation may include reconstructing a geometry of a scene into sub-elements (e.g., triangles) using depth information stored in a depth buffer. In some embodiments, the method may include algorithmic methods to generate a triangle topology for a desired image by processing the depth buffer. The algorithmic method may adaptively tesselate a scene with triangles in spaces to keep the depth data under an inputted error rate. In other embodiments, the algorithmic method may include a hierarchal tessellation process that utilizes a set of levels (e.g., Z-buffer) where each level may be an array of quads (e.g., quads containing (2″+1)×(2″+1) points). For example, FIG. **9** shows an example array of quads, according to some embodiments of this disclosure. The quads may each share an edge with the neighboring quads in the array. In further embodiments, in each quad, a number of points are considered for tessellation (e.g., generation of triangles on each quad). In some examples, the points on each quad may be read and selectively kept and/or removed during the tessellation process on each level. For example, FIG. **10** shows example quads represented by a 5-bit value. During tessellation, a flatness test may be conducted on the points to determine whether to tessellate edges based on error considerations. The hierarchical tessellation approach may enhance efficient tessellation while maintaining visual accuracy of the desired scene. In further examples, the algorithmic method may support optimization strategies including discarding triangles if their vertices have clear values. Additionally, the algorithmic methods may support odd-sized meshes and clipped objects for compatibility with diverse rendering scenarios.

[0051] In some embodiments, the algorithmic method may include hardware implementations including tile based rendering systems. The algorithmic method may include generating a data structure for each tile that may describe the triangles representing the z-surface. In other embodiments, the algorithmic method may accelerate the generation of triangles using the tile based rendering systems. For example, FIGS. **11-12** show example adaptive tessellation quads representing a z-surface. In some examples, tiles devoid of geometry may be omitted from processing. In other examples, the tile based renderer may write quad values from one to five. In some methods, level one processing may include flatness testing and utilizing z data obtained from preceding tiles (e.g., z-data from bottom side and right side boundaries). In further methods, processing at Level two may include accessing neighboring data from Level one, typically from two adjacent sides.

Example Embodiments

[0052] Example 4: A method including (1) reconstructing a geometry of a scene into sub-elements using depth information stored in a depth buffer; (2) specifying at least one parameter to selectively generate the sub-elements; and (3) accelerating the sub-element generation using a tile based renderer.

Systems and Methods for Dimensional Scene Sorting

[0053] A The present disclosure is generally directed to a scene-sorting algorithm that explicitly accounts for two-dimensional (2D) content when mixed in with a three-dimensional (3D) scene. The methods and systems disclosed herein enable for users to apply labels to 2D content to group them into virtual panels. A panel may include one or more related pieces of 2D content that lie in the same 3D plane, and users may apply a label to a piece of 2D content to indicate which panel it belongs to. The methods and systems disclosed herein may also enable users to apply a relative order (e.g., Z order, rendering order, etc.) to each piece of content in the 2D panel, which may give users selective control over the order in which each piece of content on a panel will be rendered for that panel. The selective control over the rendering order may include user inputs processed in real-time, updating the rendering order dynamically.

[0054] In one example, the disclosed systems and methods for improved dimensional scene sorting may include sorting visual content (e.g., transparent and/or non-transparent elements) within a 3D scene in a rendering order (e.g., visual content including 2D and/or 3D objects). In some methods, labels may be assigned to 2D content to form virtual panels, where the panels may include related 2D elements on the same 3D plane. On each panel, a z-order for each individual piece of 2D content may be established to provide control over their rendering order. Users may adjust the z-order of the 2D elements within a panel, determining a sequence in which the elements will be rendered. In other methods, disclosed methods (e.g., scene sorting algorithm, stable sorting algorithm, etc.) may include receiving the visual content as a list of content pieces to preserve the relative order of 2D content within each virtual panel. Further, for each virtual panel, the disclosed methods may include creating a representative boundary (e.g., rectangle) and a 3D position to encapsulate a spatial extent of related 2D content. Additionally, the representative boundary may facilitate the subsequent sorting of the visual content based on depth relationships. In some examples, the disclosed methods may include preserving the render order of the visual content to accommodate the complex mixing of 2D and 3D scenes.

[0055] In other examples, the disclosed methods may include a comparison function to determine the rendering order of the visual content by identifying the elements of the visual content. In some examples, the comparison function may prioritize non-transparent (e.g., opaque) 3D content and order 3D content based on the depth of their bounding box centers. Prioritizing non-transparent 3D elements may include assigning a higher priority value to the non-transparent 3D elements among other 3D and 2D elements. In some examples, using the comparison function may include determining the rendering order of 3D content among other 3D elements based on the depth of their respective bounding box centers. In further examples, the methods described herein may calculate depth values for the boundaries of each 2D element. The depth values may be compared to establish a rendering order that aligns with the spatial hierarchy of the scene. In some embodiments, two pieces of 2D content on the same panel may be considered identical, where the order of the 2D content may be determined based on the depth of each panel boundary center. In some embodiments, two pieces of 2D content on the same panel that are not parallel (e.g., nearly parallel) may be compared by the depth of each panel boundary center. In other embodiments, two pieces of 2D content on the same panel of 2D content that are parallel may include an overlap region between the boundaries, where the depth of each boundary at the center of the overlapping region may be compared. In some instances, the boundaries void of an overlap region may be compared using the depth of each panel boundary center.

[0056] In further embodiments, ordering the 2D content may include projecting the boundary where the visual content will be rendered into a screen space. Further, the comparison function may determine an overlap region between the 2D content in the rendered image. In some embodiments, the terms “screen space” may refer to a coordinate system of a final image and/or display output of the visual content. The comparison function may transform the visual content into the screen space to determine their depths relative to the user. By transforming the visual content into screen space, the comparison function may dynamically compare each piece of 2D and 3D content based on their respective depth relationships. In some examples, comparing the visual content based on their depth relationships may include comparing the depth of the boundary around 2D elements with one or more depths of other boundaries around 3D elements. In some examples, comparing the visual content based on their depth relationships may include comparing the depth of the boundary around 3D elements with one more depths of other boundaries around 3D elements.

[0057] The disclosed scene-sorting may integrate 2D content within a 3D scene, enabling users enhanced control over the sorting and rendering of visual elements. By allowing labels and relative order assignments to 2D content, the systems disclosed herein may facilitate the creation of virtual panels that may maintain spatial relationships and rendering priorities. This approach may enable 2D and 3D content to be mixed and displayed, preserving the intended spatial hierarchy. The implementation of the methods disclosed herein may improve the accuracy and efficiency of rendering complex scenes.

[0058] Embodiments of the present disclosure may include or be implemented in conjunction with various types of Artificial-Reality (AR) systems. AR may be any superimposed functionality and/or sensory-detectable content presented by an artificial-reality system within a user's physical surroundings. In other words, AR is a form of reality that has been adjusted in some manner before presentation to a user. AR can include and/or represent virtual reality (VR), augmented reality, mixed AR (MAR), or some combination and/or variation of these types of realities. Similarly, AR environments may include VR environments (including non-immersive, semi-immersive, and fully immersive VR environments), augmented-reality environments (including marker-based augmented-reality environments, markerless augmented-reality environments, location-based augmented-reality environments, and projection-based augmented-reality environments), hybrid-reality environments, and/or any other type or form of mixed- or alternative-reality environments.

[0059] AR content may include completely computer-generated content or computer-generated content combined with captured (e.g., real-world) content. Such AR content may include video, audio, haptic feedback, or some combination thereof, any of which may be presented in a single channel or in multiple channels (such as stereo video that produces a three-dimensional (3D) effect to the viewer). Additionally, in some embodiments, AR may also be associated with applications, products, accessories, services, or some combination thereof, that are used to, for example, create content in an artificial reality and/or are otherwise used in (e.g., to perform activities in) an artificial reality.

[0060] AR systems may be implemented in a variety of different form factors and configurations. Some AR systems may be designed to work without near-eye displays (NEDs). Other AR systems may include a NED that also provides visibility into the real world (such as, e.g., augmented-reality system **1900** in FIG. **19**) or that visually immerses a user in an artificial reality (such as, e.g., virtual-reality system **2000** in FIGS. **20**A and **20**B). While some AR devices may be self-contained systems, other AR devices may communicate and/or coordinate with external devices to provide an AR experience to a user. Examples of such external devices include handheld controllers, mobile devices, desktop computers, devices worn by a user, devices worn by one or more other users, and/or any other suitable external system.

[0061] FIGS. **13-16**B illustrate example artificial-reality (AR) systems in accordance with some embodiments. FIG. **13** shows a first AR system **1300** and first example user interactions using a wrist-wearable device **1302**, a head-wearable device (e.g., AR glasses **1900**), and/or a handheld intermediary processing device (HIPD) **1306**. FIG. **14** shows a second AR system **1400** and second example user interactions using a wrist-wearable device **1402**, AR glasses **1404**, and/or an HIPD **1406**. FIGS. **15**A and **15**B show a third AR system **1500** and third example user **1508** interactions using a wrist-wearable device **1502**, a head-wearable device (e.g., VR headset **1550**), and/or an HIPD **1506**. FIGS. **16**A and **16**B show a fourth AR system **1600** and fourth example user **1608** interactions using a wrist-wearable device **1630**, VR headset **1620**, and/or a haptic device **1660** (e.g., wearable gloves).

[0062] A wrist-wearable device **1700**, which can be used for wrist-wearable device **1302**, **1402**, **1502**, **1630**, and one or more of its components, are described below in reference to FIGS. **17** and **18**; head-wearable devices **1900** and **2000**, which can respectively be used for AR glasses **1304**, **1404** or VR headset **1550**, **1620**, and their one or more components are described below in reference to FIGS. **19-21**.

[0063] Referring to FIG. **13**, wrist-wearable device **1302**, AR glasses **1304**, and/or HIPD **1306** can communicatively couple via a network **1325** (e.g., cellular, near field, Wi-Fi, personal area network, wireless LAN, etc.). Additionally, wrist-wearable device **1302**, AR glasses **1304**, and/or HIPD **1306** can also communicatively couple with one or more servers **1330**, computers **1340** (e.g., laptops, computers, etc.), mobile devices **1350** (e.g., smartphones, tablets, etc.), and/or other electronic devices via network **1325** (e.g., cellular, near field, Wi-Fi, personal area network, wireless LAN, etc.).

[0064] In FIG. **13**, a user **1308** is shown wearing wrist-wearable device **1302** and AR glasses **1304** and having HIPD **1306** on their desk. The wrist-wearable device **1302**, AR glasses **1304**, and HIPD **1306** facilitate user interaction with an AR environment. In particular, as shown by first AR system **1300**, wrist-wearable device **1302**, AR glasses **1304**, and/or HIPD **1306** cause presentation of one or more avatars **1310**, digital representations of contacts **1312**, and virtual objects **1314**. As discussed below, user **1308** can interact with one or more avatars **1310**, digital representations of contacts **1312**, and virtual objects **1314** via wrist-wearable device **1302**, AR glasses **1304**, and/or HIPD **1306**.

[0065] User **1308** can use any of wrist-wearable device **1302**, AR glasses **1304**, and/or HIPD **1306** to provide user inputs. For example, user **1308** can perform one or more hand gestures that are detected by wrist-wearable device **1302** (e.g., using one or more EMG sensors and/or IMUs, described below in reference to FIGS. **17** and **18**) and/or AR glasses **1304** (e.g., using one or more image sensor or camera, described below in reference to FIGS. **19-10**) to provide a user input. Alternatively, or additionally, user **1308** can provide a user input via one or more touch surfaces of wrist-wearable device **1302**, AR glasses **1304**, HIPD **1306**, and/or voice commands captured by a microphone of wrist-wearable device **1302**, AR glasses **1304**, and/or HIPD **1306**. In some embodiments, wrist-wearable device **1302**, AR glasses **1304**, and/or HIPD **1306** include a digital assistant to help user **1308** in providing a user input (e.g., completing a sequence of operations, suggesting different operations or commands, providing reminders, confirming a command, etc.).

In some embodiments, user **1308** can provide a user input via one or more facial gestures and/or facial expressions. For example, cameras of wrist-wearable device **1302**, AR glasses **1304**, and/or HIPD **1306** can track eyes of user **1308** for navigating a user interface.

[0066] Wrist-wearable device **1302**, AR glasses **1304**, and/or HIPD **1306** can operate alone or in conjunction to allow user **1308** to interact with the AR environment. In some embodiments, HIPD **1306** is configured to operate as a central hub or control center for the wrist-wearable device **1302**, AR glasses **1304**, and/or another communicatively coupled device. For example, user **1308** can provide an input to interact with the AR environment at any of wrist-wearable device **1302**, AR glasses **1304**, and/or HIPD **1306**, and HIPD **1306** can identify one or more back-end and front-end tasks to cause the performance of the requested interaction and distribute instructions to cause the performance of the one or more back-end and front-end tasks at wrist-wearable device **1302**, AR glasses **1304**, and/or HIPD **1306**. In some embodiments, a back-end task is a background processing task that is not perceptible by the user (e.g., rendering content, decompression, compression, etc.), and a front-end task is a user-facing task that is perceptible to the user (e.g., presenting information to the user, providing feedback to the user, etc.). As described below, HIPD **1306** can perform the back-end tasks and provide wrist-wearable device **1302** and/or AR glasses **1304** operational data corresponding to the performed back-end tasks such that wrist-wearable device **1302** and/or AR glasses **1304** can perform the front-end tasks. In this way, HIPD **1306**, which has more computational resources and greater thermal headroom than wrist-wearable device **1302** and/or AR glasses **1304**, performs computationally intensive tasks and reduces the computer resource utilization and/or power usage of wrist-wearable device **1302** and/or AR glasses **1304**.

[0067] In the example shown by first AR system **1300**, HIPD **1306** identifies one or more back-end tasks and front-end tasks associated with a user request to initiate an AR video call with one or more other users (represented by avatar **1310** and the digital representation of contact **1312**) and distributes instructions to cause the performance of the one or more back-end tasks and front-end tasks. In particular, HIPD **1306** performs back-end tasks for processing and/or rendering image data (and other data) associated with the AR video call and provides operational data associated with the performed back-end tasks to AR glasses **1304** such that the AR glasses **1304** perform front-end tasks for presenting the AR video call (e.g., presenting avatar **1310** and digital representation of contact **1312**).

[0068] In some embodiments, HIPD **1306** can operate as a focal or anchor point for causing the presentation of information. This allows user **1308** to be generally aware of where information is presented. For example, as shown in first AR system **1300**, avatar **1310** and the digital representation of contact **1312** are presented above HIPD **1306**. In particular, HIPD **1306** and AR glasses **1304** operate in conjunction to determine a location for presenting avatar **1310** and the digital representation of contact **1312**. In some embodiments, information can be presented a predetermined distance from HIPD **1306** (e.g., within 5 meters). For example, as shown in first AR system **1300**, virtual object **1314** is presented on the desk some distance from HIPD **1306**. Similar to the above example, HIPD **1306** and AR glasses **1304** can operate in conjunction to determine a location for presenting virtual object **1314**. Alternatively, in some embodiments, presentation of information is not bound by HIPD **1306**. More specifically, avatar **1310**, digital representation of contact **1312**, and virtual object **1314** do not have to be presented within a predetermined distance of HIPD **1306**.

[0069] User inputs provided at wrist-wearable device **1302**, AR glasses **1304**, and/or HIPD **1306** are coordinated such that the user can use any device to initiate, continue, and/or complete an operation. For example, user **1308** can provide a user input to AR glasses **1304** to cause AR glasses **1304** to present virtual object **1314** and, while virtual object **1314** is presented by AR glasses **1304**, user **1308** can provide one or more hand gestures via wrist-wearable device **1302** to interact and/or manipulate virtual object **1314**.

[0070] FIG. **14** shows a user **1408** wearing a wrist-wearable device **1402** and AR glasses **1404**, and holding an HIPD **1406**. In second AR system **1400**, the wrist-wearable device **1402**, AR glasses **1404**, and/or HIPD **1406** are used to receive and/or provide one or more messages to a contact of user **1408**. In particular, wrist-wearable device **1402**, AR glasses **1404**, and/or HIPD **1406** detect and coordinate one or more user inputs to initiate a messaging application and prepare a response to a received message via the messaging application.

[0071] In some embodiments, user **1408** initiates, via a user input, an application on wrist-wearable device **1402**, AR glasses **1404**, and/or HIPD **1406** that causes the application to initiate on at least one device. For example, in second AR system **1400**, user **1408** performs a hand gesture associated with a command for initiating a messaging application (represented by messaging user interface **1416**), wrist-wearable device **1402** detects the hand gesture and, based on a determination that user **1408** is wearing AR glasses **1404**, causes AR glasses **1404** to present a messaging user interface **1416** of the messaging application. AR glasses **1404** can present messaging user interface **1416** to user **1408** via its display (e.g., as shown by a field of view **1418** of user **1408**). In some embodiments, the application is initiated and executed on the device (e.g., wrist-wearable device **1402**, AR glasses **1404**, and/or HIPD **1406**) that detects the user input to initiate the application, and the device provides another device operational data to cause the presentation of the messaging application. For example, wrist-wearable device **1402** can detect the user input to initiate a messaging application, initiate and run the messaging application, and provide operational data to AR glasses **1404** and/or HIPD **1406** to cause presentation of the messaging application. Alternatively, the application can be initiated and executed at a device other than the device that detected the user input. For example, wrist-wearable device **1402** can detect the hand gesture associated with initiating the messaging application and cause HIPD **1406** to run the messaging application and coordinate the presentation of the messaging application.

[0072] Further, user **1408** can provide a user input provided at wrist-wearable device **1402**, AR glasses **1404**, and/or HIPD **1406** to continue and/or complete an operation initiated at another device. For example, after initiating the messaging application via wrist-wearable device **1402** and while AR glasses **1404** present messaging user interface **1416**, user **1408** can provide an input at HIPD **1406** to prepare a response (e.g., shown by the swipe gesture performed on HIPD **1406**). Gestures performed by user **1408** on HIPD **1406** can be provided and/or displayed on another device. For example, a swipe gestured performed on HIPD **1406** is displayed on a virtual keyboard of messaging user interface **1416** displayed by AR glasses **1404**.

[0073] In some embodiments, wrist-wearable device **1402**, AR glasses **1404**, HIPD **1406**, and/or any other communicatively coupled device can present one or more notifications to user **1408**. The notification can be an indication of a new message, an incoming call, an application update, a status update, etc. User **1408** can select the notification via wrist-wearable device **1402**, AR glasses **1404**, and/or HIPD **1406** and can cause presentation of an application or operation associated with the notification on at least one device. For example, user **1408** can receive a notification that a message was received at wrist-wearable device **1402**, AR glasses **1404**, HIPD **1406**, and/or any other communicatively coupled device and can then provide a user input at wrist-wearable device **1402**, AR glasses **1404**, and/or HIPD **1406** to review the notification, and the device detecting the user input can cause an application associated with the notification to be initiated and/or presented at wrist-wearable device **1402**, AR glasses **1404**, and/or HIPD **1406**.

[0074] While the above example describes coordinated inputs used to interact with a messaging application, user inputs can be coordinated to interact with any number of applications including, but not limited to, gaming applications, social media applications, camera applications, web-based applications, financial applications, etc. For example, AR glasses **1404** can present to user **1408** game application data, and HIPD **1406** can be used as a controller to provide inputs to the game. Similarly, user **1408** can use wrist-wearable device **1402** to initiate a camera of AR glasses **1404**, and user **308** can use wrist-wearable device **1402**, AR glasses **1404**, and/or HIPD **1406** to manipulate the image capture (e.g., zoom in or out, apply filters, etc.) and capture image data.

[0075] Users may interact with the devices disclosed herein in a variety of ways. For example, as shown in FIGS. **15**A and **15**B, a user **1508** may interact with an AR system **1500** by donning a VR headset **1550** while holding HIPD **1506** and wearing wrist-wearable device **1502**. In this example, AR system **1500** may enable a user to interact with a game **1510** by swiping their arm. One or more of VR headset **1550**, HIPD **1506**, and wrist-wearable device **1502** may detect this gesture and, in response, may display a sword strike in game **1510**. Similarly, in FIGS. **16**A and **16**B, a user **1608** may interact with an AR system **1600** by donning a VR headset **1620** while wearing haptic device **1660** and wrist-wearable device **1630**. In this example, AR system **1600** may enable a user to interact with a game **1610** by swiping their arm. One or more of VR headset **1620**, haptic device **1660**, and wrist-wearable device **1630** may detect this gesture and, in response, may display a spell being cast in game **1510**.
[0076] Having discussed example AR systems, devices for interacting with such AR systems and other computing systems more generally will now be discussed in greater detail. Some explanations of devices and components that can be included in some or all of the example devices discussed below are explained herein for ease of reference. Certain types of the components described below may be more suitable for a particular set of devices, and less suitable for a different set of devices. But subsequent reference to the components explained here should be considered to be encompassed by the descriptions provided.
[0077] In some embodiments discussed below, example devices and systems, including electronic devices and systems, will be addressed. Such example devices and systems are not intended to be limiting, and one of skill in the art will understand that alternative devices and systems to the example devices and systems described herein may be used to perform the operations and construct the systems and devices that are described herein.
[0078] An electronic device may be a device that uses electrical energy to perform a specific function. An electronic device can be any physical object that contains electronic components such as transistors, resistors, capacitors, diodes, and integrated circuits. Examples of electronic devices include smartphones, laptops, digital cameras, televisions, gaming consoles, and music players, as well as the example electronic devices discussed herein. As described herein, an intermediary electronic device may be a device that sits between two other electronic devices and/or a subset of components of one or more electronic devices and facilitates communication, data processing, and/or data transfer between the respective electronic devices and/or electronic components.
[0079] An integrated circuit may be an electronic device made up of multiple interconnected electronic components such as transistors, resistors, and capacitors. These components may be etched onto a small piece of semiconductor material, such as silicon. Integrated circuits may include analog integrated circuits, digital integrated circuits, mixed signal integrated circuits, and/or any other suitable type or form of integrated circuit. Examples of integrated circuits include application-specific integrated circuits (ASICs), processing units, central processing units (CPUs), co-processors, and accelerators.
[0080] Analog integrated circuits, such as sensors, power management circuits, and operational amplifiers, may process continuous signals and perform analog functions such as amplification, active filtering, demodulation, and mixing. Examples of analog integrated circuits include linear integrated circuits and radio frequency circuits.
[0081] Digital integrated circuits, which may be referred to as logic integrated circuits, may include microprocessors, microcontrollers, memory chips, interfaces, power management circuits, programmable devices, and/or any other suitable type or form of integrated circuit. In some embodiments, examples of integrated circuits include central processing units (CPUs),
[0082] Processing units, such as CPUs, may be electronic components that are responsible for executing instructions and controlling the operation of an electronic device (e.g., a computer). There are various types of processors that may be used interchangeably, or may be specifically required, by embodiments described herein. For example, a processor may be: (i) a general

processor designed to perform a wide range of tasks, such as running software applications, managing operating systems, and performing arithmetic and logical operations; (ii) a microcontroller designed for specific tasks such as controlling electronic devices, sensors, and motors; (iii) an accelerator, such as a graphics processing unit (GPU), designed to accelerate the creation and rendering of images, videos, and animations (e.g., virtual-reality animations, such as three-dimensional modeling); (iv) a field-programmable gate array (FPGA) that can be programmed and reconfigured after manufacturing and/or can be customized to perform specific tasks, such as signal processing, cryptography, and machine learning; and/or (v) a digital signal processor (DSP) designed to perform mathematical operations on signals such as audio, video, and radio waves. One or more processors of one or more electronic devices may be used in various embodiments described herein.

[0083] Memory generally refers to electronic components in a computer or electronic device that store data and instructions for the processor to access and manipulate. Examples of memory can include: (i) random access memory (RAM) configured to store data and instructions temporarily; (ii) read-only memory (ROM) configured to store data and instructions permanently (e.g., one or more portions of system firmware, and/or boot loaders) and/or semi-permanently; (iii) flash memory, which can be configured to store data in electronic devices (e.g., USB drives, memory cards, and/or solid-state drives (SSDs)); and/or (iv) cache memory configured to temporarily store frequently accessed data and instructions. Memory, as described herein, can store structured data (e.g., SQL databases, MongoDB databases, GraphQL data, JSON data, etc.). Other examples of data stored in memory can include (i) profile data, including user account data, user settings, and/or other user data stored by the user, (ii) sensor data detected and/or otherwise obtained by one or more sensors, (iii) media content data including stored image data, audio data, documents, and the like, (iv) application data, which can include data collected and/or otherwise obtained and stored during use of an application, and/or any other types of data described herein.

[0084] Controllers may be electronic components that manage and coordinate the operation of other components within an electronic device (e.g., controlling inputs, processing data, and/or generating outputs). Examples of controllers can include: (i) microcontrollers, including small, low-power controllers that are commonly used in embedded systems and Internet of Things (IoT) devices; (ii) programmable logic controllers (PLCs) that may be configured to be used in industrial automation systems to control and monitor manufacturing processes; (iii) system-on-a-chip (SoC) controllers that integrate multiple components such as processors, memory, I/O interfaces, and other peripherals into a single chip; and/or (iv) DSPs.

[0085] A power system of an electronic device may be configured to convert incoming electrical power into a form that can be used to operate the device. A power system can include various components, such as (i) a power source, which can be an alternating current (AC) adapter or a direct current (DC) adapter power supply, (ii) a charger input, which can be configured to use a wired and/or wireless connection (which may be part of a peripheral interface, such as a USB, micro-USB interface, near-field magnetic coupling, magnetic inductive and magnetic resonance charging, and/or radio frequency (RF) charging), (iii) a power-management integrated circuit, configured to distribute power to various components of the device and to ensure that the device operates within safe limits (e.g., regulating voltage, controlling current flow, and/or managing heat dissipation), and/or (iv) a battery configured to store power to provide usable power to components of one or more electronic devices.

[0086] Peripheral interfaces may be electronic components (e.g., of electronic devices) that allow electronic devices to communicate with other devices or peripherals and can provide the ability to input and output data and signals. Examples of peripheral interfaces can include (i) universal serial bus (USB) and/or micro-USB interfaces configured for connecting devices to an electronic device, (ii) Bluetooth interfaces configured to allow devices to communicate with each other, including Bluetooth low energy (BLE), (iii) near field communication (NFC) interfaces configured to be short-range wireless interfaces for operations such as access control, (iv) POGO pins, which may be small, spring-loaded pins configured to provide a charging interface, (v) wireless charging interfaces, (vi) GPS interfaces, (vii) Wi-Fi interfaces for providing a connection between a device and a wireless network, and/or (viii) sensor interfaces.

[0087] Sensors may be electronic components (e.g., in and/or otherwise in electronic communication with electronic devices, such as wearable devices) configured to detect physical and environmental changes and generate electrical signals. Examples of sensors can include (i) imaging sensors for collecting imaging data (e.g., including one or more cameras disposed on a respective electronic device), (ii) biopotential-signal sensors, (iii) inertial measurement units (e.g., IMUs) for detecting, for example, angular rate, force, magnetic field, and/or changes in acceleration, (iv) heart rate sensors for measuring a user's heart rate, (v) SpO2 sensors for measuring blood oxygen saturation and/or other biometric data of a user, (vi) capacitive sensors for detecting changes in potential at a portion of a user's body (e.g., a sensor-skin interface), and/or (vii) light sensors (e.g., time-of-flight sensors, infrared light sensors, visible light sensors, etc.).

[0088] Biopotential-signal-sensing components may be devices used to measure electrical activity within the body (e.g., biopotential-signal sensors). Some types of biopotential-signal sensors include (i) electroencephalography (EEG) sensors configured to measure electrical activity in the brain to diagnose neurological disorders, (ii) electrocardiogram EKG) sensors configured to measure electrical activity of the heart to diagnose heart problems, (iii) electromyography (EMG) sensors configured to measure the electrical activity of muscles and to diagnose neuromuscular disorders, and (iv) electrooculography (EOG) sensors configure to measure the electrical activity of eye muscles to detect eye movement and diagnose eye disorders.

[0089] An application stored in memory of an electronic device (e.g., software) may include instructions stored in the memory. Examples of such applications include (i) games, (ii) word processors, (iii) messaging applications, (iv) media-streaming applications, (v) financial applications, (vi) calendars. (vii) clocks, and (viii) communication interface modules for enabling wired and/or wireless connections between different respective electronic devices (e.g., IEEE 1902.15.4, Wi-Fi, ZigBee, 6LoWPAN, Thread, Z-Wave, Bluetooth Smart, ISA100.11a, WirelessHART, or MiWi), custom or standard wired protocols (e.g., Ethernet or HomePlug), and/or any other suitable communication protocols).

[0090] A communication interface may be a mechanism that enables different systems or devices to exchange information and data with each other, including hardware, software, or a combination of both hardware and software. For example, a communication interface can refer to a physical connector and/or port on a device that enables communication with other devices (e.g., USB, Ethernet, HDMI, Bluetooth). In some embodiments, a communication interface can refer to a software layer that enables different software programs to communicate with each other (e.g., application programming interfaces (APIs), protocols like HTTP and TCP/IP, etc.).

[0091] A graphics module may be a component or software module that is designed to handle graphical operations and/or processes and can include a hardware module and/or a software module.

[0092] Non-transitory computer-readable storage media may be physical devices or storage media that can be used to store electronic data in a non-transitory form (e.g., such that the data is stored permanently until it is intentionally deleted or modified).

[0093] FIGS. **17** and **18** illustrate an example wrist-wearable device **1700** and an example computer system **1800**, in accordance with some embodiments. Wrist-wearable device **1700** is an instance of wearable device **1302** described in FIG. **13** herein, such that the wearable device **1302** should be understood to have the features of the wrist-wearable device **1700** and vice versa. FIG. **18** illustrates components of the wrist-wearable device **1700**, which can be used individually or in combination, including combinations that include other electronic devices and/or electronic components.

[0094] FIG. **17** shows a wearable band **1710** and a watch body **1720** (or capsule) being coupled, as discussed below, to form wrist-wearable device **1700**. Wrist-wearable device **1700** can perform various functions and/or operations associated with navigating through user interfaces and selectively opening applications as well as the functions and/or operations described above with reference to FIGS. **13-16**B.

[0095] As will be described in more detail below, operations executed by wrist-wearable device **1700** can include (i) presenting content to a user (e.g., displaying visual content via a display **1705**), (ii) detecting (e.g., sensing) user input (e.g., sensing a touch on peripheral button **1723** and/or at a touch screen of the display **1705**, a hand gesture detected by sensors (e.g., biopotential sensors)), (iii) sensing biometric data (e.g., neuromuscular signals, heart rate, temperature, sleep, etc.) via one or more sensors **1713**, messaging (e.g., text, speech, video, etc.); image capture via one or more imaging devices or cameras **1725**, wireless communications (e.g., cellular, near field, Wi-Fi, personal area network, etc.), location determination, financial transactions, providing haptic feedback, providing alarms, providing notifications, providing biometric authentication, providing health monitoring, providing sleep monitoring, etc.

[0096] The above-example functions can be executed independently in watch body **1720**, independently in wearable band **1710**, and/or via an electronic communication between watch body **1720** and wearable band **1710**. In some embodiments, functions can be executed on wrist-wearable device **1700** while an AR environment is being presented (e.g., via one of AR systems **1300** to **1600**). The wearable devices described herein can also be used with other types of AR environments.

[0097] Wearable band **1710** can be configured to be worn by a user such that an inner surface of a wearable structure **1711** of wearable band **1710** is in contact with the user's skin. In this example, when worn by a user, sensors **1713** may contact the user's skin. In some examples, one or more of sensors **1713** can sense biometric data such as a user's heart rate, a saturated oxygen level, temperature, sweat level, neuromuscular signals, or a combination thereof. One or more of sensors **1713** can also sense data about a user's environment including a user's motion, altitude, location, orientation, gait, acceleration, position, or a combination thereof. In some embodiment, one or more of sensors **1713** can be configured to track a position and/or motion of wearable band **1710**. One or more of sensors **1713** can include any of the sensors defined above and/or discussed below with respect to FIG. **17**.

[0098] One or more of sensors **1713** can be distributed on an inside and/or an outside surface of wearable band **1710**. In some embodiments, one or more of sensors **1713** are uniformly spaced along wearable band **1710**. Alternatively, in some embodiments, one or more of sensors **1713** are positioned at distinct points along wearable band **1710**. As shown in FIG. **17**, one or more of sensors **1713** can be the same or distinct. For example, in some embodiments, one or more of sensors **1713** can be shaped as a pill (e.g., sensor **1713***a*), an oval, a circle a square, an oblong (e.g., sensor **1713***c*) and/or any other shape that maintains contact with the user's skin (e.g., such that neuromuscular signal and/or other biometric data can be accurately measured at the user's skin). In some embodiments, one or more sensors of **1713** are aligned to form pairs of sensors (e.g., for sensing neuromuscular signals based on differential sensing within each respective sensor). For example, sensor **1713***b* may be aligned with an adjacent sensor to form sensor pair **1714***a* and sensor **1713***d* may be aligned with an adjacent sensor to form sensor pair **1714***b*. In some embodiments, wearable band **1710** does not have a sensor pair. Alternatively, in some embodiments, wearable band **1710** has a predetermined number of sensor pairs (one pair of sensors, three pairs of sensors, four pairs of sensors, six pairs of sensors, sixteen pairs of sensors, etc.).

[0099] Wearable band **1710** can include any suitable number of sensors **1713**. In some embodiments, the number and arrangement of sensors **1713** depends on the particular application for which wearable band **1710** is used. For instance, wearable band **1710** can be configured as an armband, wristband, or chest-band that include a plurality of sensors **1713** with different number of sensors **1713**, a variety of types of individual sensors with the plurality of sensors **1713**, and different arrangements for each use case, such as medical use cases as compared to gaming or general day-to-day use cases.

[0100] In accordance with some embodiments, wearable band **1710** further includes an electrical ground electrode and a shielding electrode. The electrical ground and shielding electrodes, like the sensors **1713**, can be distributed on the inside surface of the wearable band **1710** such that they contact a portion of the user's skin. For example, the electrical ground and shielding electrodes can be at an inside surface of a coupling mechanism **1716** or an inside surface of a wearable structure **1711**. The electrical ground and shielding electrodes can be formed and/or use the same components as sensors **1713**. In some embodiments, wearable band **1710** includes more than one electrical ground electrode and more than one shielding electrode.

[0101] Sensors **1713** can be formed as part of wearable structure **1711** of wearable band **1710**. In some embodiments, sensors **1713** are flush or substantially flush with wearable structure **1711** such that they do not extend beyond the surface of wearable structure **1711**. While flush with wearable structure **1711**, sensors **1713** are still configured to contact the user's skin (e.g., via a skin-contacting surface). Alternatively, in some embodiments, sensors **1713** extend beyond wearable structure **1711** a predetermined distance (e.g., 0.1-2 mm) to make contact and depress into the user's skin. In some embodiment, sensors **1713** are coupled to an actuator (not shown) configured to adjust an extension height (e.g., a distance from the surface of wearable structure **1711**) of sensors **1713** such that sensors **1713** make contact and depress into the user's skin. In some embodiments, the actuators adjust the extension height between 0.01 mm-1.2 mm. This may allow a the user to customize the positioning of sensors **1713** to improve the overall comfort of the wearable band **1710** when worn while still allowing sensors **1713** to contact the user's skin. In some embodiments, sensors **1713** are indistinguishable from wearable structure **1711** when worn by the user.

[0102] Wearable structure **1711** can be formed of an elastic material, elastomers, etc., configured to be stretched and fitted to be worn by the user. In some embodiments, wearable structure **1711** is a textile or woven fabric. As described above, sensors **1713** can be formed as part of a wearable structure **1711**. For example, sensors **1713** can be molded into the wearable structure **1711**, be integrated into a woven fabric (e.g., sensors **1713** can be sewn into the fabric and mimic the pliability of fabric and can and/or be constructed from a series woven strands of fabric).

[0103] Wearable structure **1711** can include flexible electronic connectors that interconnect sensors **1713**, the electronic circuitry, and/or other electronic components (described below in reference to FIG. **18**) that are enclosed in wearable band **1710**. In some embodiments, the flexible electronic connectors are configured to interconnect sensors **1713**, the electronic circuitry, and/or other electronic components of wearable band **1710** with respective sensors and/or other electronic components of another electronic device (e.g., watch body **1720**). The flexible electronic connectors are configured to move with wearable structure **1711** such that the user adjustment to wearable structure **1711** (e.g., resizing, pulling, folding, etc.) does not stress or strain the electrical coupling of components of wearable band **1710**.

[0104] As described above, wearable band **1710** is configured to be worn by a user. In particular, wearable band **1710** can be shaped or otherwise manipulated to be worn by a user. For example, wearable band **1710** can be shaped to have a substantially circular shape such that it can be configured to be worn on the user's lower arm or wrist. Alternatively, wearable band **1710** can be shaped to be worn on another body part of the user, such as the user's upper arm (e.g., around a bicep), forearm, chest, legs, etc. Wearable band **1710** can include a retaining mechanism **1712** (e.g., a buckle, a hook and loop fastener, etc.) for securing wearable band **1710** to the user's wrist or other body part. While wearable band **1710** is worn by the user, sensors **1713** sense data (referred to as sensor data) from the user's skin. In some examples, sensors **1713** of wearable band **1710** obtain (e.g., sense and record) neuromuscular signals.

[0105] The sensed data (e.g., sensed neuromuscular signals) can be used to detect and/or determine the user's intention to perform certain motor actions. In some examples, sensors **1713** may sense and record neuromuscular signals from the user as the user performs muscular activations (e.g., movements, gestures, etc.). The detected and/or determined motor actions (e.g., phalange (or digit) movements, wrist movements, hand movements, and/or other muscle intentions) can be used to determine control commands or control information (instructions to perform certain commands after the data is sensed) for causing a computing device to perform one or more input commands. For example, the sensed neuromuscular signals can be used to control certain user interfaces displayed on display **1705** of wrist-wearable device **1700** and/or can be transmitted to a device responsible for rendering an artificial-reality environment (e.g., a head-mounted display) to perform an action in an associated artificial-reality environment, such as to control the motion of a virtual device displayed to the user. The muscular activations performed by the user can include static gestures, such as placing the user's hand palm down on a table, dynamic gestures, such as grasping a physical or virtual object, and covert gestures that are imperceptible to another person, such as slightly tensing a joint by co-contracting opposing muscles or using sub-muscular activations. The muscular activations performed by the user can include symbolic gestures (e.g., gestures mapped to other gestures, interactions, or commands, for example, based on a gesture vocabulary that specifies the mapping of gestures to commands).

[0106] The sensor data sensed by sensors **1713** can be used to provide a user with an enhanced interaction with a physical object (e.g., devices communicatively coupled with wearable band **1710**) and/or a virtual object in an artificial-reality application generated by an artificial-reality system (e.g., user interface objects presented on the display **1705**, or another computing device (e.g., a smartphone)).

[0107] In some embodiments, wearable band **1710** includes one or more haptic devices **1846** (e.g., a vibratory haptic actuator) that are configured to provide haptic feedback (e.g., a cutaneous and/or kinesthetic sensation, etc.) to the user's skin. Sensors **1713** and/or haptic devices **1846** (shown in FIG. **18**) can be configured to operate in conjunction with multiple applications including, without limitation, health monitoring, social media, games, and artificial reality (e.g., the applications associated with artificial reality).

[0108] Wearable band **1710** can also include coupling mechanism **1716** for detachably coupling a capsule (e.g., a computing unit) or watch body **1720** (via a coupling surface of the watch body **1720**) to wearable band **1710**. For example, a cradle or a shape of coupling mechanism **1716** can correspond to shape of watch body **1720** of wrist-wearable device **1700**. In particular, coupling mechanism **1716** can be configured to receive a coupling surface proximate to the bottom side of watch body **1720** (e.g., a side opposite to a front side of watch body **1720** where display **1705** is located), such that a user can push watch body **1720** downward into coupling mechanism **1716** to attach watch body **1720** to coupling mechanism **1716**. In some embodiments, coupling mechanism **1716** can be configured to receive a top side of the watch body **1720** (e.g., a side proximate to the front side of watch body **1720** where display **1705** is located) that is pushed upward into the cradle, as opposed to being pushed downward into coupling mechanism **1716**. In some embodiments, coupling mechanism **1716** is an integrated component of wearable band **1710** such that wearable band **1710** and coupling mechanism **1716** are a single unitary structure. In some embodiments, coupling mechanism **1716** is a type of frame or shell that allows watch body **1720** coupling surface to be retained within or on wearable band **1710** coupling mechanism **1716** (e.g., a cradle, a tracker band, a support base, a clasp, etc.).

[0109] Coupling mechanism **1716** can allow for watch body **1720** to be detachably coupled to the wearable band **1710** through a friction fit, magnetic coupling, a rotation-based connector, a shear-pin coupler, a retention spring, one or more magnets, a clip, a pin shaft, a hook and loop fastener, or a combination thereof. A user can perform any type of motion to couple the watch body **1720** to wearable band **1710** and to decouple the watch body **1720** from the wearable band **1710**. For example, a user can twist, slide, turn, push, pull, or rotate watch body **1720** relative to wearable band **1710**, or a combination thereof, to attach watch body **1720** to wearable band **1710** and to detach watch body **1720** from wearable band **1710**. Alternatively, as discussed below, in some embodiments, the watch body **1720** can be decoupled from the wearable band **1710** by actuation of a release mechanism **1729**.

[0110] Wearable band **1710** can be coupled with watch body **1720** to increase the functionality of wearable band **1710** (e.g., converting wearable band **1710** into wrist-wearable device **1700**, adding an additional computing unit and/or battery to increase computational resources and/or a battery life of wearable band **1710**, adding additional sensors to improve sensed data, etc.). As described above, wearable band **1710** and coupling mechanism **1716** are configured to operate independently (e.g., execute functions independently) from watch body **1720**. For example, coupling mechanism **1716** can include one or more sensors **1713** that contact a user's skin when wearable band **1710** is worn by the user, with or without watch body **1720** and can provide sensor data for determining control commands.

[0111] A user can detach watch body **1720** from wearable band **1710** to reduce the encumbrance of wrist-wearable device **1700** to the user. For embodiments in which watch body **1720** is removable, watch body **1720** can be referred to as a removable structure, such that in these embodiments wrist-wearable device **1700** includes a wearable portion (e.g., wearable band **1710**) and a removable structure (e.g., watch body **1720**).

[0112] Turning to watch body **1720**, in some examples watch body **1720** can have a substantially rectangular or circular shape. Watch body **1720** is configured to be worn by the user on their wrist or on another body part. More specifically, watch body **1720** is sized to be easily carried by the user, attached on a portion of the user's clothing, and/or coupled to wearable band **1710** (forming the wrist-wearable device **1700**). As described above, watch body **1720** can have a shape corresponding to coupling mechanism **1716** of wearable band **1710**. In some embodiments, watch body **1720** includes a single release mechanism **1729** or multiple release mechanisms (e.g., two release mechanisms **1729** positioned on opposing sides of watch body **1720**, such as spring-loaded buttons) for decoupling watch body **1720** from wearable band **1710**. Release mechanism **1729** can include, without limitation, a button, a knob, a plunger, a handle, a lever, a fastener, a clasp, a dial, a latch, or a combination thereof.

[0113] A user can actuate release mechanism **1729** by pushing, turning, lifting, depressing, shifting, or performing other actions on release mechanism **1729**. Actuation of release mechanism **1729** can release (e.g., decouple) watch body **1720** from coupling mechanism **1716** of wearable band **1710**, allowing the user to use watch body **1720** independently from wearable band **1710** and vice versa. For example, decoupling watch body **1720** from wearable band **1710** can allow a user to capture images using rear-facing camera **1725***b*. Although release mechanism **1729** is shown positioned at a corner of watch body **1720**, release mechanism **1729** can be positioned anywhere on watch body **1720** that is convenient for the user to actuate. In addition, in some embodiments, wearable band **1710** can also include a respective release mechanism for decoupling watch body **1720** from coupling mechanism **1716**. In some embodiments, release mechanism **1729** is optional and watch body **1720** can be decoupled from coupling mechanism **1716** as described above (e.g., via twisting, rotating, etc.).

[0114] Watch body **1720** can include one or more peripheral buttons **1723** and **1727** for performing various operations at watch body **1720**. For example, peripheral buttons **1723** and **1727** can be used to turn on or wake (e.g., transition from a sleep state to an active state) display **1705**, unlock watch body **1720**, increase or decrease a volume, increase or decrease a brightness, interact with one or more applications, interact with one or more user interfaces, etc. Additionally or alternatively, in some embodiments, display **1705** operates as a touch screen and allows the user to provide one or more inputs for interacting with watch body **1720**.

[0115] In some embodiments, watch body **1720** includes one or more sensors **1721**. Sensors **1721** of watch body **1720** can be the same or distinct from sensors **1713** of wearable band **1710**. Sensors **1721** of watch body **1720** can be distributed on an inside and/or an outside surface of watch body **1720**. In some embodiments, sensors **1721** are configured to contact a user's skin when watch body **1720** is worn by the user. For example, sensors **1721** can be placed on the bottom side of watch body **1720** and coupling mechanism **1716** can be a cradle with an opening that allows the bottom side of watch body **1720** to directly contact the user's skin. Alternatively, in some embodiments, watch body **1720** does not include sensors that are configured to contact the user's skin (e.g., including sensors internal and/or external to the watch body **1720** that are configured to sense data of watch body **1720** and the surrounding environment). In some embodiments, sensors **1721** are configured to track a position and/or motion of watch body **1720**.
[0116] Watch body **1720** and wearable band **1710** can share data using a wired communication method (e.g., a Universal Asynchronous Receiver/Transmitter (UART), a USB transceiver, etc.) and/or a wireless communication method (e.g., near field communication, Bluetooth, etc.). For example, watch body **1720** and wearable band **1710** can share data sensed by sensors **1713** and **1721**, as well as application and device specific information (e.g., active and/or available applications, output devices (e.g., displays, speakers, etc.), input devices (e.g., touch screens, microphones, imaging sensors, etc.).
[0117] In some embodiments, watch body **1720** can include, without limitation, a front-facing camera **1725***a* and/or a rear-facing camera **1725***b*, sensors **1721** (e.g., a biometric sensor, an IMU, a heart rate sensor, a saturated oxygen sensor, a neuromuscular signal sensor, an altimeter sensor, a temperature sensor, a bioimpedance sensor, a pedometer sensor, an optical sensor (e.g., imaging sensor **1863**), a touch sensor, a sweat sensor, etc.). In some embodiments, watch body **1720** can include one or more haptic devices **1876** (e.g., a vibratory haptic actuator) that is configured to provide haptic feedback (e.g., a cutaneous and/or kinesthetic sensation, etc.) to the user. Sensors **1821** and/or haptic device **1876** can also be configured to operate in conjunction with multiple applications including, without limitation, health monitoring applications, social media applications, game applications, and artificial reality applications (e.g., the applications associated with artificial reality).
[0118] As described above, watch body **1720** and wearable band **1710**, when coupled, can form wrist-wearable device **1700**. When coupled, watch body **1720** and wearable band **1710** may operate as a single device to execute functions (operations, detections, communications, etc.) described herein. In some embodiments, each device may be provided with particular instructions for performing the one or more operations of wrist-wearable device **1700**. For example, in accordance with a determination that watch body **1720** does not include neuromuscular signal sensors, wearable band **1710** can include alternative instructions for performing associated instructions (e.g., providing sensed neuromuscular signal data to watch body **1720** via a different electronic device). Operations of wrist-wearable device **1700** can be performed by watch body **1720** alone or in conjunction with wearable band **1710** (e.g., via respective processors and/or hardware components) and vice versa. In some embodiments, operations of wrist-wearable device **1700**, watch body **1720**, and/or wearable band **1710** can be performed in conjunction with one or more processors and/or hardware components.
[0119] As described below with reference to the block diagram of FIG. **18**, wearable band **1710** and/or watch body **1720** can each include independent resources required to independently execute functions. For example, wearable band **1710** and/or watch body **1720** can each include a power source (e.g., a battery), a memory, data storage, a processor (e.g., a central processing unit (CPU)), communications, a light source, and/or input/output devices.
[0120] FIG. **18** shows block diagrams of a computing system **1830** corresponding to wearable band **1710** and a computing system **1860** corresponding to watch body **1720** according to some embodiments. Computing system **1800** of wrist-wearable device **1700** may include a combination

of components of wearable band computing system **1830** and watch body computing system **1860**, in accordance with some embodiments.

[0121] Watch body **1720** and/or wearable band **1710** can include one or more components shown in watch body computing system **1860**. In some embodiments, a single integrated circuit may include all or a substantial portion of the components of watch body computing system **1860** included in a single integrated circuit. Alternatively, in some embodiments, components of the watch body computing system **1860** may be included in a plurality of integrated circuits that are communicatively coupled. In some embodiments, watch body computing system **1860** may be configured to couple (e.g., via a wired or wireless connection) with wearable band computing system **1830**, which may allow the computing systems to share components, distribute tasks, and/or perform other operations described herein (individually or as a single device).

[0122] Watch body computing system **1860** can include one or more processors **1879**, a controller **1877**, a peripherals interface **1861**, a power system **1895**, and memory (e.g., a memory **1880**).

[0123] Power system **1895** can include a charger input **1896**, a power-management integrated circuit (PMIC) **1897**, and a battery **1898**. In some embodiments, a watch body **1720** and a wearable band **1710** can have respective batteries (e.g., battery **1898** and **1859**) and can share power with each other. Watch body **1720** and wearable band **1710** can receive a charge using a variety of techniques. In some embodiments, watch body **1720** and wearable band **1710** can use a wired charging assembly (e.g., power cords) to receive the charge. Alternatively, or in addition, watch body **1720** and/or wearable band **1710** can be configured for wireless charging. For example, a portable charging device can be designed to mate with a portion of watch body **1720** and/or wearable band **1710** and wirelessly deliver usable power to battery **1898** of watch body **1720** and/or battery **1859** of wearable band **1710**. Watch body **1720** and wearable band **1710** can have independent power systems (e.g., power system **1895** and **1856**, respectively) to enable each to operate independently. Watch body **1720** and wearable band **1710** can also share power (e.g., one can charge the other) via respective PMICs (e.g., PMICs **1897** and **1858**) and charger inputs (e.g., **1857** and **1896**) that can share power over power and ground conductors and/or over wireless charging antennas.

[0124] In some embodiments, peripherals interface **1861** can include one or more sensors **1821**. Sensors **1821** can include one or more coupling sensors **1862** for detecting when watch body **1720** is coupled with another electronic device (e.g., a wearable band **1710**). Sensors **1821** can include one or more imaging sensors **1863** (e.g., one or more of cameras **1825**, and/or separate imaging sensors **1863** (e.g., thermal-imaging sensors)). In some embodiments, sensors **1821** can include one or more SpO2 sensors **1864**. In some embodiments, sensors **1821** can include one or more biopotential-signal sensors (e.g., EMG sensors **1865**, which may be disposed on an interior, user-facing portion of watch body **1720** and/or wearable band **1710**). In some embodiments, sensors **1821** may include one or more capacitive sensors **1866**. In some embodiments, sensors **1821** may include one or more heart rate sensors **1867**. In some embodiments, sensors **1821** may include one or more IMU sensors **1868**. In some embodiments, one or more IMU sensors **1868** can be configured to detect movement of a user's hand or other location where watch body **1720** is placed or held.

[0125] In some embodiments, one or more of sensors **1821** may provide an example human-machine interface. For example, a set of neuromuscular sensors, such as EMG sensors **1865**, may be arranged circumferentially around wearable band **1710** with an interior surface of EMG sensors **1865** being configured to contact a user's skin. Any suitable number of neuromuscular sensors may be used (e.g., between 2 and 20 sensors). The number and arrangement of neuromuscular sensors may depend on the particular application for which the wearable device is used. For example, wearable band **1710** can be used to generate control information for controlling an augmented reality system, a robot, controlling a vehicle, scrolling through text, controlling a virtual avatar, or any other suitable control task.

[0126] In some embodiments, neuromuscular sensors may be coupled together using flexible electronics incorporated into the wireless device, and the output of one or more of the sensing components can be optionally processed using hardware signal processing circuitry (e.g., to perform amplification, filtering, and/or rectification). In other embodiments, at least some signal processing of the output of the sensing components can be performed in software such as processors **1879**. Thus, signal processing of signals sampled by the sensors can be performed in hardware, software, or by any suitable combination of hardware and software, as aspects of the technology described herein are not limited in this respect.
[0127] Neuromuscular signals may be processed in a variety of ways. For example, the output of EMG sensors **1865** may be provided to an analog front end, which may be configured to perform analog processing (e.g., amplification, noise reduction, filtering, etc.) on the recorded signals. The processed analog signals may then be provided to an analog-to-digital converter, which may convert the analog signals to digital signals that can be processed by one or more computer processors. Furthermore, although this example is as discussed in the context of interfaces with EMG sensors, the embodiments described herein can also be implemented in wearable interfaces with other types of sensors including, but not limited to, mechanomyography (MMG) sensors, sonomyography (SMG) sensors, and electrical impedance tomography (EIT) sensors.
[0128] In some embodiments, peripherals interface **1861** includes a near-field communication (NFC) component **1869**, a global-position system (GPS) component **1870**, a long-term evolution (LTE) component **1871**, and/or a Wi-Fi and/or Bluetooth communication component **1872**. In some embodiments, peripherals interface **1861** includes one or more buttons **1873** (e.g., peripheral buttons **1723** and **1727** in FIG. **17**), which, when selected by a user, cause operation to be performed at watch body **1720**. In some embodiments, the peripherals interface **1861** includes one or more indicators, such as a light emitting diode (LED), to provide a user with visual indicators (e.g., message received, low battery, active microphone and/or camera, etc.).
[0129] Watch body **1720** can include at least one display **1705** for displaying visual representations of information or data to a user, including user-interface elements and/or three-dimensional virtual objects. The display can also include a touch screen for inputting user inputs, such as touch gestures, swipe gestures, and the like. Watch body **1720** can include at least one speaker **1874** and at least one microphone **1875** for providing audio signals to the user and receiving audio input from the user. The user can provide user inputs through microphone **1875** and can also receive audio output from speaker **1874** as part of a haptic event provided by haptic controller **1878**. Watch body **1720** can include at least one camera **1825**, including a front camera **1825***a* and a rear camera **1825***b*. Cameras **1825** can include ultra-wide-angle cameras, wide angle cameras, fish-eye cameras, spherical cameras, telephoto cameras, depth-sensing cameras, or other types of cameras.
[0130] Watch body computing system **1860** can include one or more haptic controllers **1878** and associated componentry (e.g., haptic devices **1876**) for providing haptic events at watch body **1720** (e.g., a vibrating sensation or audio output in response to an event at the watch body **1720**). Haptic controllers **1878** can communicate with one or more haptic devices **1876**, such as electroacoustic devices, including a speaker of the one or more speakers **1874** and/or other audio components and/or electromechanical devices that convert energy into linear motion such as a motor, solenoid, electroactive polymer, piezoelectric actuator, electrostatic actuator, or other tactile output generating components (e.g., a component that converts electrical signals into tactile outputs on the device). Haptic controller **1878** can provide haptic events to that are capable of being sensed by a user of watch body **1720**. In some embodiments, one or more haptic controllers **1878** can receive input signals from an application of applications **1882**.
[0131] In some embodiments, wearable band computing system **1830** and/or watch body computing system **1860** can include memory **1880**, which can be controlled by one or more memory controllers of controllers **1877**. In some embodiments, software components stored in memory **1880** include one or more applications **1882** configured to perform operations at the watch

body **1720**. In some embodiments, one or more applications **1882** may include games, word processors, messaging applications, calling applications, web browsers, social media applications, media streaming applications, financial applications, calendars, clocks, etc. In some embodiments, software components stored in memory **1880** include one or more communication interface modules **1883** as defined above. In some embodiments, software components stored in memory **1880** include one or more graphics modules **1884** for rendering, encoding, and/or decoding audio and/or visual data and one or more data management modules **1885** for collecting, organizing, and/or providing access to data **1887** stored in memory **1880**. In some embodiments, one or more of applications **1882** and/or one or more modules can work in conjunction with one another to perform various tasks at the watch body **1720**.

[0132] In some embodiments, software components stored in memory **1880** can include one or more operating systems **1881** (e.g., a Linux-based operating system, an Android operating system, etc.). Memory **1880** can also include data **1887**. Data **1887** can include profile data **1888**A, sensor data **1889**A, media content data **1890**, and application data **1891**.

[0133] It should be appreciated that watch body computing system **1860** is an example of a computing system within watch body **1720**, and that watch body **1720** can have more or fewer components than shown in watch body computing system **1860**, can combine two or more components, and/or can have a different configuration and/or arrangement of the components. The various components shown in watch body computing system **1860** are implemented in hardware, software, firmware, or a combination thereof, including one or more signal processing and/or application-specific integrated circuits.

[0134] Turning to the wearable band computing system **1830**, one or more components that can be included in wearable band **1710** are shown. Wearable band computing system **1830** can include more or fewer components than shown in watch body computing system **1860**, can combine two or more components, and/or can have a different configuration and/or arrangement of some or all of the components. In some embodiments, all, or a substantial portion of the components of wearable band computing system **1830** are included in a single integrated circuit. Alternatively, in some embodiments, components of wearable band computing system **1830** are included in a plurality of integrated circuits that are communicatively coupled. As described above, in some embodiments, wearable band computing system **1830** is configured to couple (e.g., via a wired or wireless connection) with watch body computing system **1860**, which allows the computing systems to share components, distribute tasks, and/or perform other operations described herein (individually or as a single device).

[0135] Wearable band computing system **1830**, similar to watch body computing system **1860**, can include one or more processors **1849**, one or more controllers **1847** (including one or more haptics controllers **1848**), a peripherals interface **1831** that can includes one or more sensors **1813** and other peripheral devices, a power source (e.g., a power system **1856**), and memory (e.g., a memory **1850**) that includes an operating system (e.g., an operating system **1851**), data (e.g., data **1854** including profile data **1888**B, sensor data **1889**B, etc.), and one or more modules (e.g., a communications interface module **1852**, a data management module **1853**, etc.).

[0136] One or more of sensors **1813** can be analogous to sensors **1821** of watch body computing system **1860**. For example, sensors **1813** can include one or more coupling sensors **1832**, one or more SpO2 sensors **1834**, one or more EMG sensors **1835**, one or more capacitive sensors **1836**, one or more heart rate sensors **1837**, and one or more IMU sensors **1838**.

[0137] Peripherals interface **1831** can also include other components analogous to those included in peripherals interface **1861** of watch body computing system **1860**, including an NFC component **1839**, a GPS component **1840**, an LTE component **1841**, a Wi-Fi and/or Bluetooth communication component **1842**, and/or one or more haptic devices **1846** as described above in reference to peripherals interface **1861**. In some embodiments, peripherals interface **1831** includes one or more buttons **1843**, a display **1833**, a speaker **1844**, a microphone **1845**, and a camera **1855**. In some

embodiments, peripherals interface **1831** includes one or more indicators, such as an LED.
[0138] It should be appreciated that wearable band computing system **1830** is an example of a computing system within wearable band **1710**, and that wearable band **1710** can have more or fewer components than shown in wearable band computing system **1830**, combine two or more components, and/or have a different configuration and/or arrangement of the components. The various components shown in wearable band computing system **1830** can be implemented in one or more of a combination of hardware, software, or firmware, including one or more signal processing and/or application-specific integrated circuits.
[0139] Wrist-wearable device **1700** with respect to FIG. **17** is an example of wearable band **1710** and watch body **1720** coupled together, so wrist-wearable device **1700** will be understood to include the components shown and described for wearable band computing system **1830** and watch body computing system **1860**. In some embodiments, wrist-wearable device **1700** has a split architecture (e.g., a split mechanical architecture, a split electrical architecture, etc.) between watch body **1720** and wearable band **1710**. In other words, all of the components shown in wearable band computing system **1830** and watch body computing system **1860** can be housed or otherwise disposed in a combined wrist-wearable device **1700** or within individual components of watch body **1720**, wearable band **1710**, and/or portions thereof (e.g., a coupling mechanism **1716** of wearable band **1710**).
[0140] The techniques described above can be used with any device for sensing neuromuscular signals but could also be used with other types of wearable devices for sensing neuromuscular signals (such as body-wearable or head-wearable devices that might have neuromuscular sensors closer to the brain or spinal column).
[0141] In some embodiments, wrist-wearable device **1700** can be used in conjunction with a head-wearable device (e.g., AR glasses **1900** and VR system **2010**) and wrist-wearable device **1700** can also be configured to be used to allow a user to control any aspect of the artificial reality (e.g., by using EMG-based gestures to control user interface objects in the artificial reality and/or by allowing a user to interact with the touchscreen on the wrist-wearable device to also control aspects of the artificial reality). Having thus described example wrist-wearable devices, attention will now be turned to example head-wearable devices, such AR glasses **1900** and VR headset **2010**.
[0142] FIGS. **19** to **21** show example artificial-reality systems, which can be used as or in connection with wrist-wearable device **1700**. In some embodiments, AR system **1900** includes an eyewear device **1902**, as shown in FIG. **19**. In some embodiments, VR system **2010** includes a head-mounted display (HMD) **2012**, as shown in FIGS. **20**A and **20**B. In some embodiments, AR system **1900** and VR system **2010** can include one or more analogous components (e.g., components for presenting interactive artificial-reality environments, such as processors, memory, and/or presentation devices, including one or more displays and/or one or more waveguides), some of which are described in more detail with respect to FIG. **21**. As described herein, a head-wearable device can include components of eyewear device **1902** and/or head-mounted display **2012**. Some embodiments of head-wearable devices do not include any displays, including any of the displays described with respect to AR system **1900** and/or VR system **2010**. While the example artificial-reality systems are respectively described herein as AR system **1900** and VR system **2010**, either or both of the example AR systems described herein can be configured to present fully-immersive virtual-reality scenes presented in substantially all of a user's field of view or subtler augmented-reality scenes that are presented within a portion, less than all, of the user's field of view.
[0143] FIG. **19** show an example visual depiction of AR system **1900**, including an eyewear device **1902** (which may also be described herein as augmented-reality glasses, and/or smart glasses). AR system **1900** can include additional electronic components that are not shown in FIG. **19**, such as a wearable accessory device and/or an intermediary processing device, in electronic communication or otherwise configured to be used in conjunction with the eyewear device **1902**. In some embodiments, the wearable accessory device and/or the intermediary processing device may be

configured to couple with eyewear device **1902** via a coupling mechanism in electronic communication with a coupling sensor **2124** (FIG. **21**), where coupling sensor **2124** can detect when an electronic device becomes physically or electronically coupled with eyewear device **1902**. In some embodiments, eyewear device **1902** can be configured to couple to a housing **2190** (FIG. **21**), which may include one or more additional coupling mechanisms configured to couple with additional accessory devices. The components shown in FIG. **19** can be implemented in hardware, software, firmware, or a combination thereof, including one or more signal-processing components and/or application-specific integrated circuits (ASICs).

[0144] Eyewear device **1902** includes mechanical glasses components, including a frame **1904** configured to hold one or more lenses (e.g., one or both lenses **1906-1** and **1906-2**). One of ordinary skill in the art will appreciate that eyewear device **1902** can include additional mechanical components, such as hinges configured to allow portions of frame **1904** of eyewear device **1902** to be folded and unfolded, a bridge configured to span the gap between lenses **1906-1** and **1906-2** and rest on the user's nose, nose pads configured to rest on the bridge of the nose and provide support for eyewear device **1902**, earpieces configured to rest on the user's ears and provide additional support for eyewear device **1902**, temple arms configured to extend from the hinges to the earpieces of eyewear device **1902**, and the like. One of ordinary skill in the art will further appreciate that some examples of AR system **1900** can include none of the mechanical components described herein. For example, smart contact lenses configured to present artificial reality to users may not include any components of eyewear device **1902**.

[0145] Eyewear device **1902** includes electronic components, many of which will be described in more detail below with respect to FIG. **10**. Some example electronic components are illustrated in FIG. **19**, including acoustic sensors **1925-1**, **1925-2**, **1925-3**, **1925-4**, **1925-5**, and **1925-6**, which can be distributed along a substantial portion of the frame **1904** of eyewear device **1902**. Eyewear device **1902** also includes a left camera **1939**A and a right camera **1939**B, which are located on different sides of the frame **1904**. Eyewear device **1902** also includes a processor **1948** (or any other suitable type or form of integrated circuit) that is embedded into a portion of the frame **1904**.

[0146] FIGS. **20**A and **20**B show a VR system **2010** that includes a head-mounted display (HMD) **2012** (e.g., also referred to herein as an artificial-reality headset, a head-wearable device, a VR headset, etc.), in accordance with some embodiments. As noted, some artificial-reality systems (e.g., AR system **1900**) may, instead of blending an artificial reality with actual reality, substantially replace one or more of a user's visual and/or other sensory perceptions of the real world with a virtual experience (e.g., AR systems **1500** and **1600**).

[0147] HMD **2012** includes a front body **2014** and a frame **2016** (e.g., a strap or band) shaped to fit around a user's head. In some embodiments, front body **2014** and/or frame **2016** include one or more electronic elements for facilitating presentation of and/or interactions with an AR and/or VR system (e.g., displays, IMUs, tracking emitter or detectors). In some embodiments, HMD **2012** includes output audio transducers (e.g., an audio transducer **2018**), as shown in FIG. **20**B. In some embodiments, one or more components, such as the output audio transducer(s) **2018** and frame **2016**, can be configured to attach and detach (e.g., are detachably attachable) to HMD **2012** (e.g., a portion or all of frame **2016**, and/or audio transducer **2018**), as shown in FIG. **20**B. In some embodiments, coupling a detachable component to HMD **2012** causes the detachable component to come into electronic communication with HMD **2012**.

[0148] FIGS. **20**A and **20**B also show that VR system **2010** includes one or more cameras, such as left camera **2039**A and right camera **2039**B, which can be analogous to left and right cameras **1939**A and **1939**B on frame **1904** of eyewear device **1902**. In some embodiments, VR system **2010** includes one or more additional cameras (e.g., cameras **2039**C and **2039**D), which can be configured to augment image data obtained by left and right cameras **2039**A and **2039**B by providing more information. For example, camera **2039**C can be used to supply color information that is not discerned by cameras **2039**A and **2039**B. In some embodiments, one or more of cameras

2039A to **2039**D can include an optional IR cut filter configured to remove IR light from being received at the respective camera sensors.

[0149] FIG. **21** illustrates a computing system **2120** and an optional housing **2190**, each of which show components that can be included in AR system **1900** and/or VR system **2010**. In some embodiments, more or fewer components can be included in optional housing **2190** depending on practical restraints of the respective AR system being described.

[0150] In some embodiments, computing system **2120** can include one or more peripherals interfaces **2122**A and/or optional housing **2190** can include one or more peripherals interfaces **2122**B. Each of computing system **2120** and optional housing **2190** can also include one or more power systems **2142**A and **2142**B, one or more controllers **2146** (including one or more haptic controllers **2147**), one or more processors **2148**A and **2148**B (as defined above, including any of the examples provided), and memory **2150**A and **2150**B, which can all be in electronic communication with each other. For example, the one or more processors **2148**A and **2148**B can be configured to execute instructions stored in memory **2150**A and **2150**B, which can cause a controller of one or more of controllers **2146** to cause operations to be performed at one or more peripheral devices connected to peripherals interface **2122**A and/or **2122**B. In some embodiments, each operation described can be powered by electrical power provided by power system **2142**A and/or **2142**B.

[0151] In some embodiments, peripherals interface **2122**A can include one or more devices configured to be part of computing system **2120**, some of which have been defined above and/or described with respect to the wrist-wearable devices shown in FIGS. **17** and **18**. For example, peripherals interface **2122**A can include one or more sensors **2123**A. Some example sensors **2123**A include one or more coupling sensors **2124**, one or more acoustic sensors **2125**, one or more imaging sensors **2126**, one or more EMG sensors **2127**, one or more capacitive sensors **2128**, one or more IMU sensors **2129**, and/or any other types of sensors explained above or described with respect to any other embodiments discussed herein.

[0152] In some embodiments, peripherals interfaces **2122**A and **2122**B can include one or more additional peripheral devices, including one or more NFC devices **2130**, one or more GPS devices **2131**, one or more LTE devices **2132**, one or more Wi-Fi and/or Bluetooth devices **2133**, one or more buttons **2134** (e.g., including buttons that are slidable or otherwise adjustable), one or more displays **2135**A and **2135**B, one or more speakers **2136**A and **2136**B, one or more microphones **2137**, one or more cameras **2138**A and **2138**B (e.g., including the left camera **2139**A and/or a right camera **2139**B), one or more haptic devices **2140**, and/or any other types of peripheral devices defined above or described with respect to any other embodiments discussed herein.

[0153] AR systems can include a variety of types of visual feedback mechanisms (e.g., presentation devices). For example, display devices in AR system **1900** and/or VR system **2010** can include one or more liquid-crystal displays (LCDs), light emitting diode (LED) displays, organic LED (OLED) displays, and/or any other suitable types of display screens. Artificial-reality systems can include a single display screen (e.g., configured to be seen by both eyes), and/or can provide separate display screens for each eye, which can allow for additional flexibility for varifocal adjustments and/or for correcting a refractive error associated with a user's vision. Some embodiments of AR systems also include optical subsystems having one or more lenses (e.g., conventional concave or convex lenses, Fresnel lenses, or adjustable liquid lenses) through which a user can view a display screen.

[0154] For example, respective displays **2135**A and **2135**B can be coupled to each of the lenses **1906-1** and **1906-2** of AR system **1900**. Displays **2135**A and **2135**B may be coupled to each of lenses **1906-1** and **1906-2**, which can act together or independently to present an image or series of images to a user. In some embodiments, AR system **1900** includes a single display **2135**A or **2135**B (e.g., a near-eye display) or more than two displays **2135**A and **2135**B. In some embodiments, a first set of one or more displays **2135**A and **2135**B can be used to present an augmented-reality environment, and a second set of one or more display devices **2135**A and **2135**B can be used to present a virtual-reality environment. In some embodiments, one or more waveguides are used in conjunction with presenting artificial-reality content to the user of AR system **1900** (e.g., as a means of delivering light from one or more displays **2135**A and **2135**B to the user's eyes). In some embodiments, one or more waveguides are fully or partially integrated into the eyewear device **1902**. Additionally, or alternatively to display screens, some artificial-reality systems include one or more projection systems. For example, display devices in AR system **1900** and/or VR system **2010** can include micro-LED projectors that project light (e.g., using a waveguide) into display devices, such as clear combiner lenses that allow ambient light to pass through. The display devices can refract the projected light toward a user's pupil and can enable a user to simultaneously view both artificial-reality content and the real world. Artificial-reality systems can also be configured with any other suitable type or form of image projection system. In some embodiments, one or more waveguides are provided additionally or alternatively to the one or more display(s) **2135**A and **2135**B.

[0155] Computing system **2120** and/or optional housing **2190** of AR system **1900** or VR system **2010** can include some or all of the components of a power system **2142**A and **2142**B. Power systems **2142**A and **2142**B can include one or more charger inputs **2143**, one or more PMICs **2144**, and/or one or more batteries **2145**A and **2144**B.

[0156] Memory **2150**A and **2150**B may include instructions and data, some or all of which may be stored as non-transitory computer-readable storage media within the memories **2150**A and **2150**B. For example, memory **2150**A and **2150**B can include one or more operating systems **2151**, one or more applications **2152**, one or more communication interface applications **2153**A and **2153**B, one or more graphics applications **2154**A and **2154**B, one or more AR processing applications **2155**A and **2155**B, and/or any other types of data defined above or described with respect to any other embodiments discussed herein.

[0157] Memory **2150**A and **2150**B also include data **2160**A and **2160**B, which can be used in conjunction with one or more of the applications discussed above. Data **2160**A and **2160**B can include profile data **2161**, sensor data **2162**A and **2162**B, media content data **2163**A, AR application data **2164**A and **2164**B, and/or any other types of data defined above or described with respect to any other embodiments discussed herein.

[0158] In some embodiments, controller **2146** of eyewear device **1902** may process information generated by sensors **2123**A and/or **2123**B on eyewear device **1902** and/or another electronic device within AR system **1900**. For example, controller **2146** can process information from acoustic sensors **1925-1** and **1925-2**. For each detected sound, controller **2146** can perform a direction of arrival (DOA) estimation to estimate a direction from which the detected sound arrived at eyewear device **1902** of R system **1900**. As one or more of acoustic sensors **2125** (e.g., the acoustic sensors **1925-1**, **1925-2**) detects sounds, controller **2146** can populate an audio data set with the information (e.g., represented in FIG. **10** as sensor data **2162**A and **2162**B).

[0159] In some embodiments, a physical electronic connector can convey information between eyewear device **1902** and another electronic device and/or between one or more processors **1948**, **2148**A, **2148**B of AR system **1900** or VR system **2010** and controller **2146**. The information can be in the form of optical data, electrical data, wireless data, or any other transmittable data form. Moving the processing of information generated by eyewear device **1902** to an intermediary processing device can reduce weight and heat in the eyewear device, making it more comfortable and safer for a user. In some embodiments, an optional wearable accessory device (e.g., an electronic neckband) is coupled to eyewear device **1902** via one or more connectors. The connectors can be wired or wireless connectors and can include electrical and/or non-electrical (e.g., structural) components. In some embodiments, eyewear device **1902** and the wearable accessory device can operate independently without any wired or wireless connection between them.

[0160] In some situations, pairing external devices, such as an intermediary processing device (e.g., HIPD **1306**, **1406**, **1506**) with eyewear device **1902** (e.g., as part of AR system **1900**) enables eyewear device **1902** to achieve a similar form factor of a pair of glasses while still providing sufficient battery and computation power for expanded capabilities. Some, or all, of the battery power, computational resources, and/or additional features of AR system **1900** can be provided by a paired device or shared between a paired device and eyewear device **1902**, thus reducing the weight, heat profile, and form factor of eyewear device **1902** overall while allowing eyewear device **1902** to retain its desired functionality. For example, the wearable accessory device can allow components that would otherwise be included on eyewear device **1902** to be included in the wearable accessory device and/or intermediary processing device, thereby shifting a weight load from the user's head and neck to one or more other portions of the user's body. In some embodiments, the intermediary processing device has a larger surface area over which to diffuse and disperse heat to the ambient environment. Thus, the intermediary processing device can allow for greater battery and computation capacity than might otherwise have been possible on eyewear device **1902** standing alone. Because weight carried in the wearable accessory device can be less invasive to a user than weight carried in the eyewear device **1902**, a user may tolerate wearing a lighter eyewear device and carrying or wearing the paired device for greater lengths of time than the user would tolerate wearing a heavier eyewear device standing alone, thereby enabling an artificial-reality environment to be incorporated more fully into a user's day-to-day activities.

[0161] AR systems can include various types of computer vision components and subsystems. For example, AR system **1900** and/or VR system **2010** can include one or more optical sensors such as two-dimensional (2D) or three-dimensional (3D) cameras, time-of-flight depth sensors, structured light transmitters and detectors, single-beam or sweeping laser rangefinders, 3D LiDAR sensors, and/or any other suitable type or form of optical sensor. An AR system can process data from one or more of these sensors to identify a location of a user and/or aspects of the use's real-world physical surroundings, including the locations of real-world objects within the real-world physical surroundings. In some embodiments, the methods described herein are used to map the real world, to provide a user with context about real-world surroundings, and/or to generate digital twins (e.g., interactable virtual objects), among a variety of other functions. For example, FIGS. **20**A and **20**B show VR system **2010** having cameras **2039**A to **2039**D, which can be used to provide depth information for creating a voxel field and a two-dimensional mesh to provide object information to the user to avoid collisions.

[0162] In some embodiments, AR system **1900** and/or VR system **2010** can include haptic (tactile) feedback systems, which may be incorporated into headwear, gloves, body suits, handheld controllers, environmental devices (e.g., chairs or floormats), and/or any other type of device or system, such as the wearable devices discussed herein. The haptic feedback systems may provide various types of cutaneous feedback, including vibration, force, traction, shear, texture, and/or temperature. The haptic feedback systems may also provide various types of kinesthetic feedback, such as motion and compliance. The haptic feedback may be implemented using motors, piezoelectric actuators, fluidic systems, and/or a variety of other types of feedback mechanisms. The haptic feedback systems may be implemented independently of other artificial-reality devices, within other artificial-reality devices, and/or in conjunction with other artificial-reality devices.

[0163] In some embodiments of an artificial reality system, such as AR system **1900** and/or VR system **2010**, ambient light (e.g., a live feed of the surrounding environment that a user would normally see) can be passed through a display element of a respective head-wearable device presenting aspects of the AR system. In some embodiments, ambient light can be passed through a portion less that is less than all of an AR environment presented within a user's field of view (e.g., a portion of the AR environment co-located with a physical object in the user's real-world environment that is within a designated boundary (e.g., a guardian boundary) configured to be used by the user while they are interacting with the AR environment). For example, a visual user interface element (e.g., a notification user interface element) can be presented at the head-wearable device, and an amount of ambient light (e.g., 15-50% of the ambient light) can be passed through the user interface element such that the user can distinguish at least a portion of the physical environment over which the user interface element is being displayed.

Example Embodiments

[0164] Example 5: A method including (1) sorting visual content within a three-dimensional (3D) scene in a rendering order by: for each set of two-dimensional (2D) elements within a same virtual plane: creating a boundary around the 2D elements; using the boundary to set the rendering order of each element with the set of 2D elements; and rendering the visual content based on the rendering order.

[0165] Example 6: The method of example 5, where sorting the visual content in the rendering order further includes, for each set of 2D elements, prioritizing non-transparent 3D elements among other 3D elements and 2D elements within the three-dimensional scene.

[0166] Example 7: The method of claim **5**, where sorting the visual content in the rendering order further includes, for each set of 2D elements, projecting the boundary where the visual content will be rendered into a screen space.

[0167] Example 8: The method of claim **5**, where using the boundary to set the rendering order of each element with the set of 2D elements further includes comparing a depth of the boundary around the 2D elements with one or more depths of other boundaries.

[0168] Example 9: The method of claim **8**, where using the boundary to set the rendering order of each element with the set of 2D elements further includes comparing the depth of the boundary around the 2D elements with the one or more depths of the other boundaries around 3D elements.

[0169] Example 10: The method of claim **8**, where using the boundary to set the rendering order of each element with the set of 2D elements further includes comparing the depth of the boundary around 3D elements with the one or more depths of the other boundaries around the 3D elements.

[0170] Example 11: The method of claim **5**, where rendering the visual content within the same virtual plane further includes enabling a user to apply selective control over the rendering order in which the 2D elements will be rendered.

Claims

1. A method comprising: sensing a temperature value of a battery circuit; activating a heat dissipation element within the battery circuit when the temperature value reaches a threshold; discharging heat from the battery circuit via the activated heat dissipation element; and deactivating the heat dissipation element when the temperature value falls below the threshold.

2. A head-mounted display comprising: a display unit configured to display computer-generated imagery to a user; and a housing that retains the display unit; wherein, when the head-mounted display is mounted on a head of the user and the display unit is positioned in a top portion of a field of view of the user: the display unit obstructs the top portion of the field of view of the user; and the housing is dimensioned to provide the user with a substantially unobstructed bottom portion of the field of view of a real-world environment to the user.

3. A computer-implemented method of augmenting image data, the method comprising: retrieving a subset of image data from a private database in response to a user input; privatizing the subset of image data; and generating at least one image based on the privatized subset of image data and the user input.

4. A method comprising: reconstructing a geometry of a scene into sub-elements using depth information stored in a depth buffer; specifying at least one parameter to selectively generate the sub-elements; and accelerating the sub-element generation using a tile based renderer.

5. A method comprising: sorting visual content within a three-dimensional (3D) scene in a rendering order by: for each set of two-dimensional (2D) elements within a same virtual plane: creating a boundary around the 2D elements; using the boundary to set the rendering order of each

element with the set of 2D elements; and rendering the visual content based on the rendering order.
6. The method of claim 5, wherein sorting the visual content in the rendering order further comprises, for each set of 2D elements, prioritizing non-transparent 3D elements among other 3D elements and 2D elements within the three-dimensional scene.
7. The method of claim 5, wherein sorting the visual content in the rendering order further comprises, for each set of 2D elements, projecting the boundary where the visual content will be rendered into a screen space.
8. The method of claim 5, wherein using the boundary to set the rendering order of each element with the set of 2D elements further comprises comparing a depth of the boundary around the 2D elements with one or more depths of other boundaries.
9. The method of claim 8, wherein using the boundary to set the rendering order of each element with the set of 2D elements further comprises comparing the depth of the boundary around the 2D elements with the one or more depths of the other boundaries around 3D elements.
10. The method of claim 8, wherein using the boundary to set the rendering order of each element with the set of 2D elements further comprises comparing the depth of the boundary around 3D elements with the one or more depths of the other boundaries around the 3D elements.
11. The method of claim 5, wherein rendering the visual content within the same virtual plane further comprises enabling a user to apply selective control over the rendering order in which the 2D elements will be rendered.